

Real-time Data Integration & Observability

Target group: SAP HANA Tech-Con
Aim: Introduction of the Data Integrity Firewall

Author
Kush Kashyap, Ashish Yadav, Jacob Pribil

22 July 2026



Kush Kashyap

Architecture & Integration Lead



Ashish Yadav

SAP BI Architect



Jacob Pribil

Data Engineer

Carl Zeiss AG

About the company



Headquarters in Oberkochen, Germany

- ZEISS was founded in **1846** by optician **Carl Zeiss** in **Jena** and the current headquarters is in **Oberkochen**, Germany.
- Carl Zeiss along with physicist **Ernst Abbe**, and glass chemist **Otto Schott**, laid the foundation for the current multinational optics and precision manufacturing company.
- **46,600+** employees globally
- Today, with **180 years of history**, ZEISS continues this legacy of technological excellence on a global scale.

Shaping the future

The ZEISS segments



Consumer Markets



Semiconductor Manufacturing Technology



Industrial Quality & Research



Medical Technology



ZEISS possess a variety of business segments generating a multitude of data sets per second.

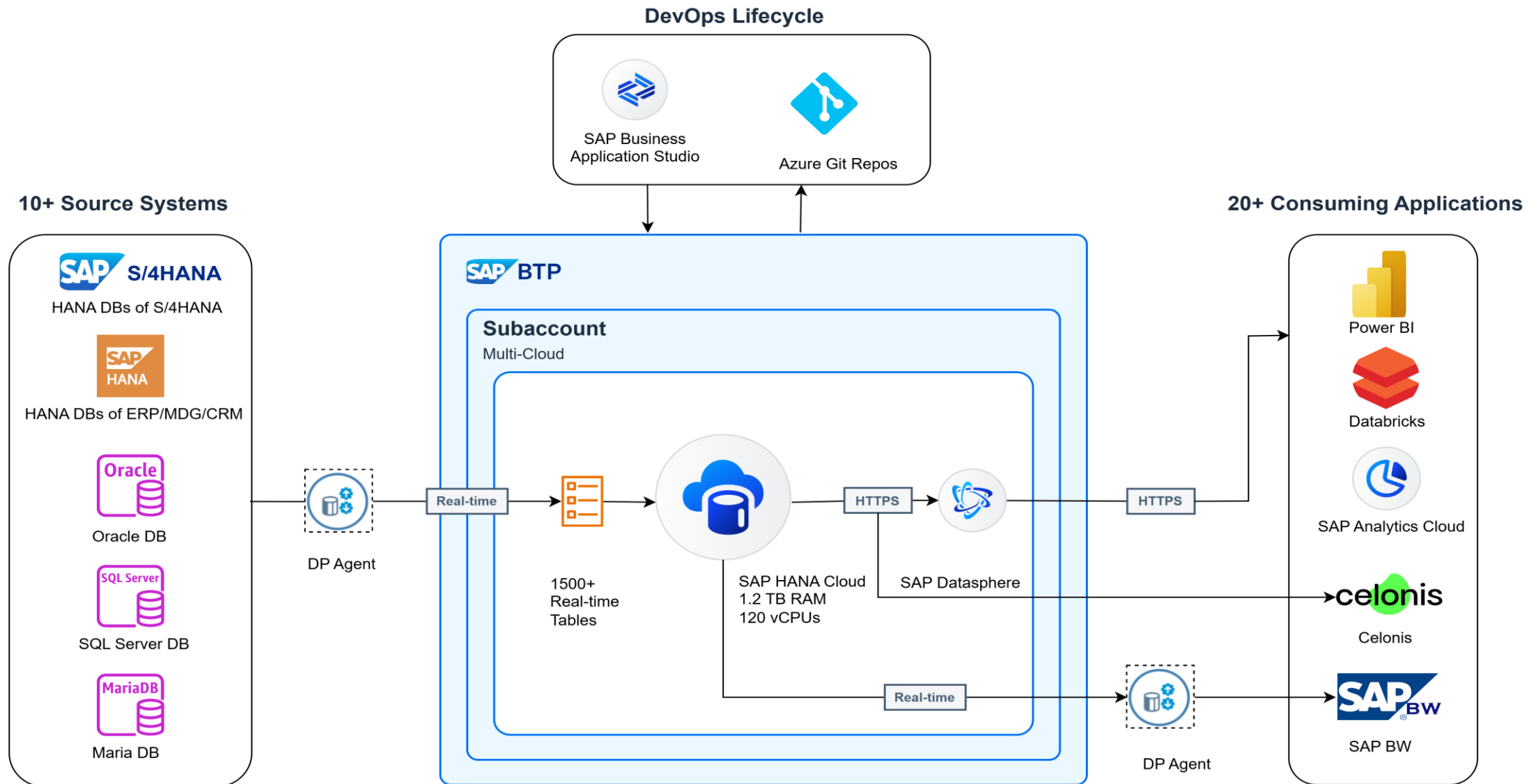
WHO WE ARE

The **SAP Data Platform Team** is part of ZEISS Digital Partners at ZEISS and is centered around **core data foundation and BI topics.**

WHAT WE DO

We are building a data foundation for enterprise data which is **consistent, integrated and usable** across systems and consumers.

HANA Cloud @ ZEISS Architecture Diagram



HANA Cloud @ ZEISS

Overview



Real-Time Enterprise Data Integration – Replicate data from SAP and non-SAP source systems into SAP HANA Cloud using SAP Smart Data Integration (SDI).

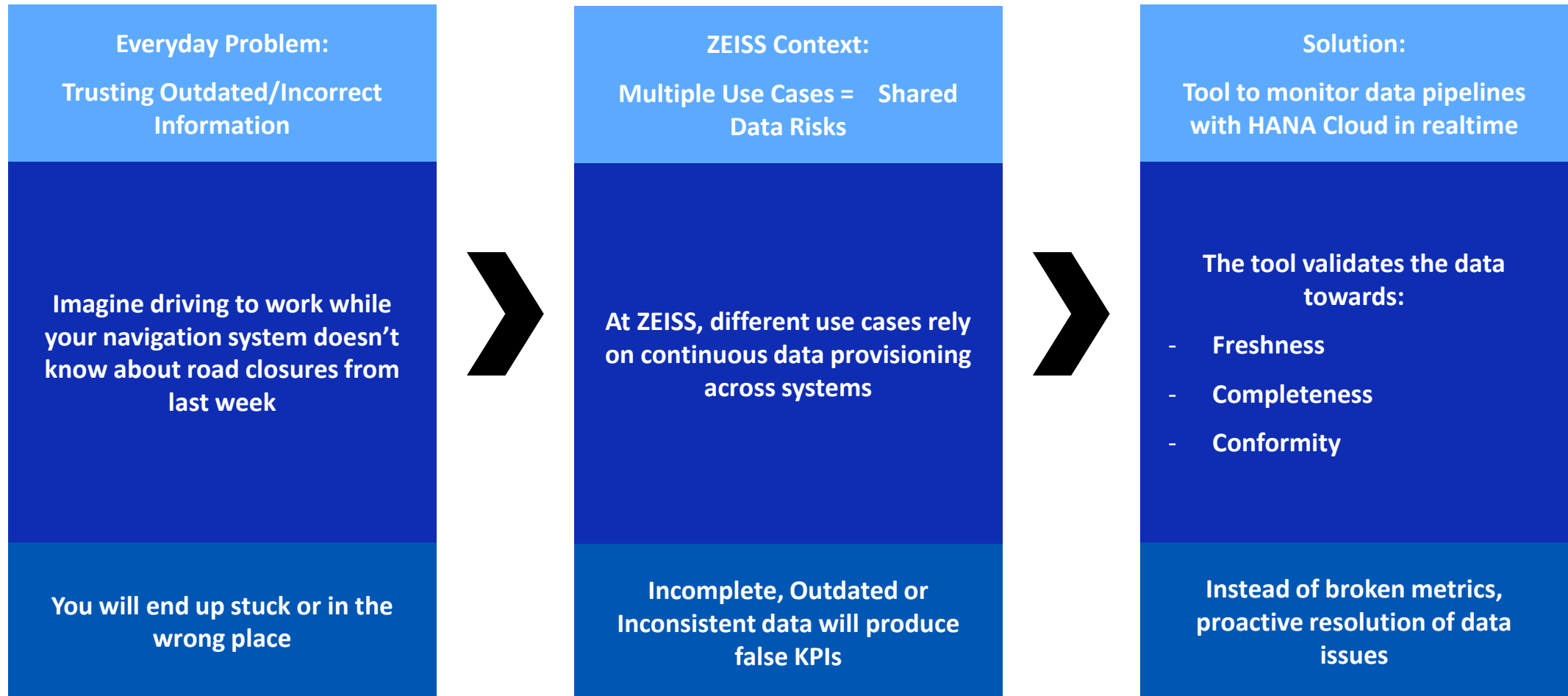
Business-Ready Data in SAP Datasphere – Federate, transform, and model centralized data into trusted semantic views for enterprise analytics and reporting.

Optimized Delta Processing – Leverage UTC timestamp-based incremental loading to ensure efficient synchronization and eliminate unnecessary full data reloads.

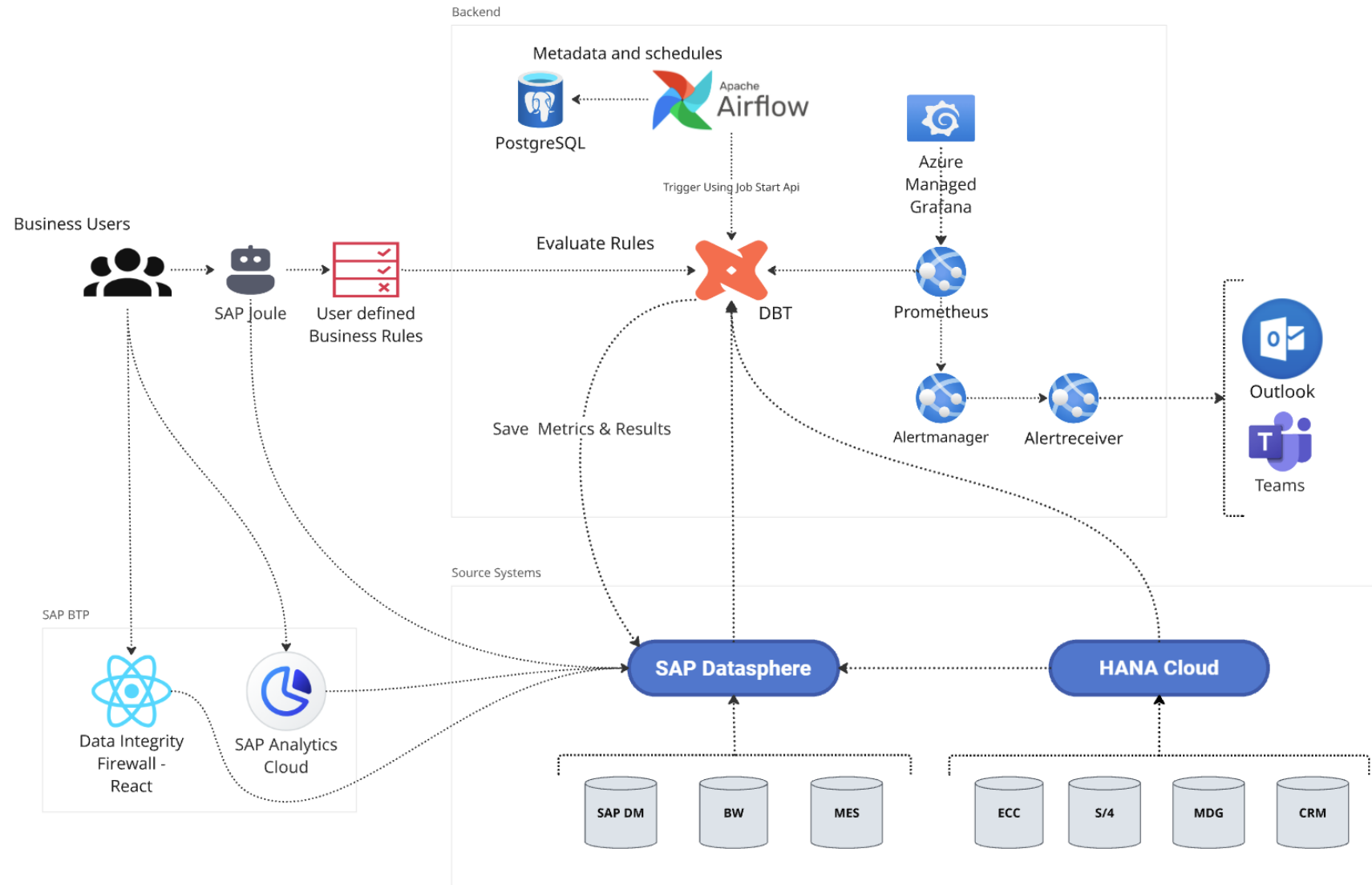
Proactive Data Observability – Monitor data freshness, replication health, and pipeline integrity to ensure reliable, high-quality, real-time business data.

Data Integrity @ ZEISS

From Data Risk to Data Integrity



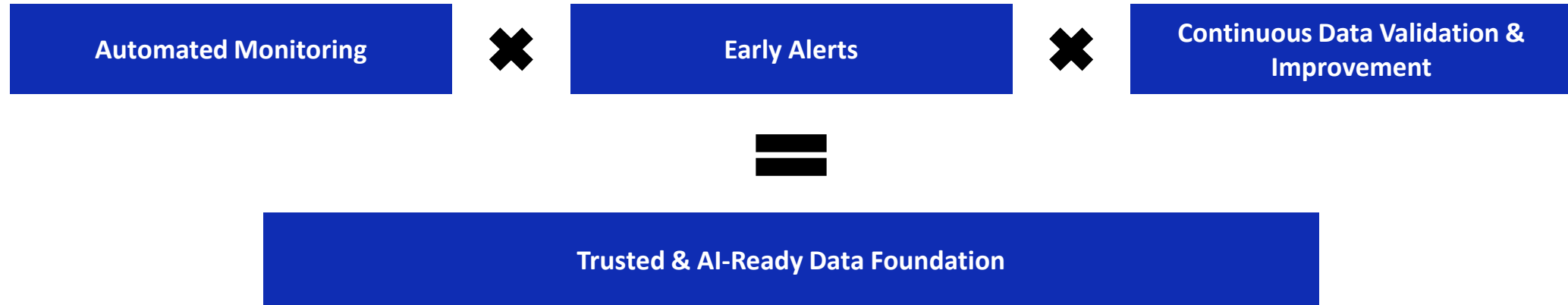
Data Integrity @ ZEISS Architecture



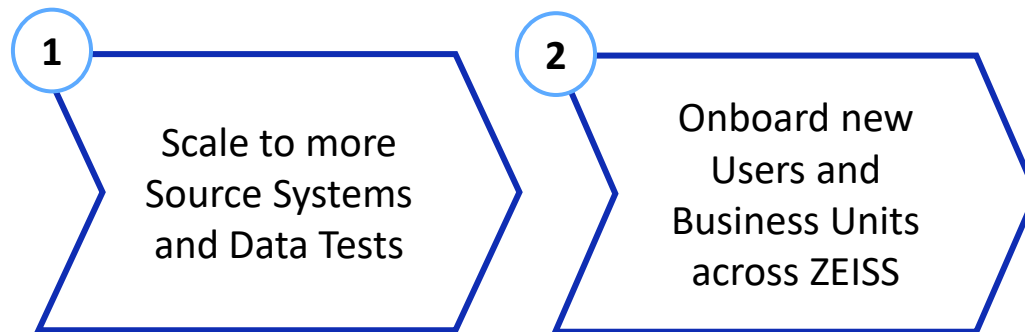
Live Demo

Data Integrity @ ZEISS

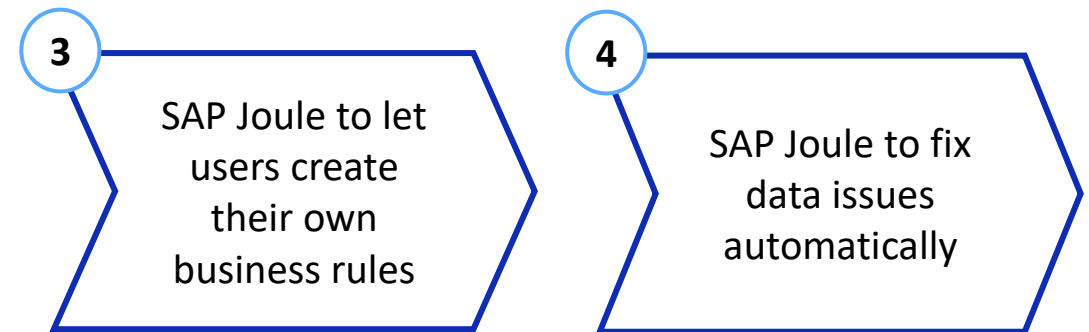
Impact & Outlook



Quick Wins



Mid-Term Innovations





Seeing beyond